Computer Programming 1

Laboratory 07

(functions and procedures - part 4)

Ex: Count Vowels and Consonants

- Write a function which takes as argument a character **chr** and returns true if it is a wovel. (Variation: return an enum, VOWEL, CONSONANT)
- Write a program which keeps asking the user for a char and counts the number of vowels and consonants entered by the user. Have the program stop when the user enters a number

Types of implementations:

1) By using functions of the cctype library (e.g., isalnum, isalpha, islowwer, tolower, etc.)
2) Without using functions in the cctype library (e.g., isalnum, isalpha, islowwer, tolower, etc.)

Ex: Math functions

- **Factorial:**
  - The factorial of a number is defined as N * (N – 1) * 1
  - Write a function which computes the factorial of a number
- **Max3**
  - Write a function which returns the highest of three numbers, passed as argument
- **Infinite Series**
  - Write a function which takes as input a base B, a number N and computes the geometric series
  - (Sum of 1/B^p, where p is between 1 and N)

$$\sum_{i=1}^{N} \frac{1}{B^i}$$

Ex: From Physics

- **Gravity:**
  - The distance **d** run by an object falling on Earth is given by the formula

$$d = \frac{1}{2}gt^2$$

Where g is 9.80665 m/s2.

Write a function which takes as input the time t and computes the distance run by an object.

Use the function above to print to standard output the distance run at increasing time units (e.g., after 1sec distance is ..., after 2secs, …). Stop when distance is > 100m or another costant of your liking.

Variation: ask the user the interval between two consecutive time steps.

Ex: From Physics

- **Speed of light:**
  - Light travels at 299792458 m/s (called "**c**") and the formula to compute the distance travelled is **d = c * t (and, viceversa, t = d /c)**
  - Compute the time it takes for light to reach a planet of our solar system, given the distances on the table on the right (e.g, about 8 minutes to reach Earth, ….)
  - Radio communication travels at the speed of light. Compute how long it takes to transmit data from a planet to another (e.g., from Mars to Earth)
  - Compose your solution with the following functions:
    - A function which returns the time, given the distance
    - A function which converts from seconds to minutes and seconds (you can also write two functions, the first returning the minutes and the second the seconds)
    - A function which computes the distance between two points in space (e.g., the distance between Mars and Earth, given the distance of both from the Sun)
    - Ask data to the user (either numerically … Enter the distance or with a switch statement and characters identifying the planets)

Sun/Planet	Distance from the Sun (km)
Sun	0
Mercury	57910000
Venus	108200000
Earth	149600000
Mars	227940000
Jupiter	778330000
Saturn	1429400000
Uranus	2870990000
Neptune	4504000000
Pluto (Dwarf Planet)	5900000000